Technical appendices

Verifier-guided tool planning for evidence-grounded engineering decisions with compact language models

Appendices A–C provide evaluation rules, experimental settings, supporting results, execution specifications and tool relationships. Labels match the accompanying manuscript. References to numbered main-text equations use the revised manuscript numbering.

Appendix A Evaluation and Supervision Details

This appendix specifies evaluation units, terminal decisions, metric aggregation and trajectory supervision.

A.1 Evaluation units and terminal decisions

The base task is the diagnostic evaluation unit. Prompt rewrites, repeated decoding and counterfactual variants are grouped with the same task and registered source group $\Gamma(e)$, preserving their shared physical data. The primary evaluation includes 240 of 249 candidate tasks. Nine were excluded for corrupt files or unparseable critical metadata using rules fixed before execution. The task specification requires diagnosis for 222 tasks and abstention or a request for essential information for 18. All methods share these exclusions and sealed labels.

Table A.1. Run-level EDS gates.

Gate	Name	Run-level pass condition
D	Correct terminal decision	The final diagnosis agrees with the sealed ground truth at the required granularity, or the system abstains correctly on a task for which the specification requires abstention.
X	Actual execution	All cited tool calls were actually executed and produced logged observations; no environmental output is fabricated, recomputed, or substituted by the model.
P	Prerequisite compliance	Every dispatched call passes offline checks of parameter schema, physical units, metadata, applicability domain and tool-specific prerequisites. Any violation makes $P = 0$ for that raw execution, whether or not the call is cited in the final report. Blocked proposals are not dispatched calls.
T	Trajectory validity	The action–observation sequence conforms to required upstream states, dependency partial order, budget, timeout, error-recovery, and terminal rules, and does not access privileged labels.
G	Sufficient evidence	Evidence required for the target diagnostic granularity is complete, and critical claims pass provenance, numerical, semantic, and dependency validation.

The offline evaluator applies all five gates in Eq. (19). Any failed gate yields $\mathrm{EDS} = 0$; timeouts, budget exhaustion, execution errors and missing valid terminal states remain failures. The aggregation rules below distinguish run-level outcomes, task-level majorities and item-level ratios.

Table A.2. Terminal decisions and failure handling.

Outcome	Condition	Recorded result
Accepted diagnosis	The target-granularity evidence gate passes and $c_e \geq \theta^*$.	Diagnosis, atomic claims and evidence ledger.
Downgraded diagnosis	A coarser granularity permitted by the request passes its recomputed evidence gate and score threshold.	Coarser diagnosis; not an abstention.
Evidential abstention	No permitted candidate passes the evidence gate because required information, claim validity, granularity constraints or a relevant high-priority conflict remains unresolved.	$\perp_{\mathrm{evid}}$, with confirmed evidence and gaps.

Outcome	Condition	Recorded result
Selective abstention	A permitted candidate passes its evidence gate but its frozen acceptance score is below the threshold.	$\perp_{\text{select}}$, recorded separately.
Repair	Validation fails recoverably while time or budget remains.	Structured feedback to the planner.
Budget failure	Budget exhaustion or timeout occurs before a valid terminal state.	$\perp_{\text{budget}}$; EDS $= 0$; retain in denominator.
Execution failure	An unrecoverable tool, parser or runtime error occurs.	$\perp_{\text{exec}}$; EDS $= 0$; retain in denominator.

A.2 Metric aggregation

For method m, seed s and task e, task-level EDS is 1 if more than half of the prompt/repeat runs succeed, and 0 otherwise (ties fail). The primary estimate weights tasks and seeds equally. Table A.3 defines the other metrics.

$$\bar{Y}_{se}(m) = \mathrm{I}\left\{\frac{1}{n_{V,e}n_{R,e}}\sum_{v\in V_e}\sum_{r\in R_e} Y_{sevr}(m) > \frac{1}{2}\right\}$$

$$n_{V,e} = |V_e|, \qquad n_{R,e} = |R_e|.$$

$$\hat{\theta}_m = \frac{1}{S}\sum_{s=1}^{S}\left[\frac{1}{|E|}\sum_{e\in E}\bar{Y}_{se}(m)\right].$$

I{·} denotes the indicator; untrained methods use $S = 1$. Source grouping retains within-group dependence for intervals (Appendix B.3); point estimates weight tasks equally.

Table A.3. Aggregation and denominator rules.

Metric family	Within-task / within-seed rule	Across-task rule
EDS	A task–seed outcome is 1 only when more than half of its prompt/repeat runs have EDS $= 1$; ties are 0. SQV has one run per task/seed/pipeline.	Equal-weight base tasks, followed by equal-weight training seeds.
Ratio metrics: TCPR, PC, NF, REC, and UCR	Retain applicable item-level numerators and denominators. TCPR audits actually dispatched calls; blocked proposals contribute no call item. Failed runs retain their dispatched calls.	Sum applicable numerators and denominators across runs and seeds. Do not average task percentages or apply task-majority aggregation to call compliance.
Macro-F1	Derive one task-level terminal class under the frozen hierarchical aggregation rule; on answer-required tasks, abstention or no valid terminal state is a false negative.	Compute classwise F1 over eligible tasks and macro-average classes.
Abstention classification, coverage, and unsafe acceptance	Use a strict majority across prompt/repeat terminal types and labels within each task and seed. No strict majority is failure; it is not a legal abstention.	For Table 3, compute coverage / 240 and unsafe acceptance / 18 within each seed, then average seeds.
Selective risk and AURC	Average the calibrated acceptance score within each base task.	Sweep frozen scores without threshold retuning; retain the hard evidence gate and preserve task and source grouping.
Pass^3	A prompt–task pair passes only when all three repeated decodings have EDS $= 1$.	Average over prompt–task pairs, then training seeds.

Pass^3 is the proportion of prompt–task pairs whose three repeated decodings all satisfy EDS. Each seed contains $180 \times 3 = 540$ sealed-test pairs and $60 \times 2 = 120$ HB pairs: 660 per seed, or 3,300 for a five-seed method. Pooling equally sized seed denominators is equivalent to averaging seed rates. The observed all-success counts for

the reference methods are 414/660 for Fixed expert, 506/660 for Oracle, 231/660 for Unadapted, 1,551/3,300 for Final-label-only, 1,690/3,300 for D-only-full, 342/660 for Dynamic rule, 356/660 for Cost-efficient frontier and 425/660 for High-capability frontier. VG-MTP has 2,026 all-success triplets across five seeds, giving $\mathrm{Pass}^3 = 2{,}026/3{,}300 = 61.39\%$.

For the primary EDS endpoint, nine raw runs per sealed-test task require at least five successes, and six runs per HB task require at least four. Failed and timed-out runs remain in the planned denominator. D-only-full and Dynamic rule have task-equal raw EDS of 68.75% and 67.50%, with pooled successes of 6,630/9,900 and 1,302/1,980. Task failure events and Pass^3 retain their own aggregation rules.

VG-MTP and LLM-process-full have 7,188 and 7,038 successes among 9,900 raw runs each, 894 and 873 successes among 1,200 task–seed pairs, and task-equal raw rates of 74.25% and 72.67%. The corresponding pooled raw rates are 72.61% and 71.09%. For VG-MTP, sealed/HB raw successes are 5,526/8,100 and 1,662/1,800, with majority counts 594/900 and 300/300. For LLM, these are 5,418/8,100 and 1,620/1,800, with majority counts 583/900 and 290/300. All rates retain failures and preserve the registered task weights.

For VG-MTP, coverage is $1{,}068/1{,}200 = 89.00\%$ and unsafe acceptance is $12/90 = 13.33\%$; the corresponding LLM-process-full counts are $1{,}074/1{,}200 = 89.50\%$ and $15/90 = 16.67\%$. These terminal-outcome rates are separate from strict-majority EDS. Failure-event incidence in Repository Table S14 follows its task-level definition.

Ratio metrics pool applicable item counts across runs and seeds. A zero denominator is recorded as NA and contributes no items: PC and UCR require evidence-bearing claims, NF requires numerical claims, and TCPR requires applicable dispatched calls. A no-tool abstention therefore contributes no call items. REC uses the fixed requested-granularity evidence set, while a permitted downgraded diagnosis is gated against its own candidate evidence set. REC is 1 when the task specification requires no tool evidence. On answer-required tasks, abstention or failure is a false negative for the true class in Macro-F1.

All 84 SQV tasks require diagnosis and have one prompt and one execution per seed. EDS is averaged across tasks within each seed and then across five seeds. Macro-F1 uses the fixed normal, inner-race and outer-race classes within each seed, followed by seed averaging. Abstentions, unrecognized outputs and missing valid terminal states count as false negatives for the true class. Overall Macro-F1 is recomputed over all 84 tasks within each seed. The primary evaluation retains its separate denominators.

Claim validation checks execution provenance, numerical values, units, semantic support and dependencies. Each citation must resolve to an executed output that supports the associated claim. A separate offline implementation recomputes UCR from raw outputs, and blinded experts assess residual semantic errors.

A.3 Trajectory supervision and runtime checks

The primary supervision comparison uses a common pool of 1,280 candidate trajectories and 1,024 trajectories in each final corpus. VG-MTP requires deterministic $D/X/P/T/G$ approval. LLM-process-full requires D and unanimous approval from the fixed LLM panel. D-only-full requires D, and Final-label-only uses a terminal-label projection. Matching covers tasks, physical training sources, fault classes, speed regimes, path lengths, teacher sources and corpus size. Full-trajectory conditions share targets, context format, 192 optimizer updates and packed-sequence capacity. Nominal corpus tokens and actual consumed tokens are recorded separately (Appendix B.9).

Training loss covers structured planner calls, visible repair decisions and terminal reports. System instructions, task payloads, actual tool observations and verifier observations remain context; these positions and padding are masked from the loss. Teacher-private reasoning is excluded from prediction targets.

Before dispatch, the runtime verifier checks schema, units, prerequisites, dependency order and budget against the current state. Blocked proposals return repair feedback without consuming dispatch slots. Execution records enter the ledger, and valid observations support claims. Repair decisions and skipped dependencies remain in the planner context to prevent repeated unchanged proposals. A permitted coarser diagnosis must pass its own evidence gate and frozen score threshold, while REC retains the original requested granularity.

The acceptance map and operating threshold are calibrated on development data and fixed before test evaluation. Test-time sweeps assess risk and coverage while retaining the hard evidence gate.

Appendix B Experimental Settings and Supplementary Results

B.1 Data sources and partition scope

Table B.1. Data composition and evaluation roles.

Source	Signals and public composition	Conditions	Use and interpretation
Huang–Baddour v2 (Huang, 2019)	Vibration and encoder reference; 60 records at 200 kHz.	Five health states × four speed trajectories × three trials.	External evaluation: 60 tasks, grouped by five health states for the main paired analysis.
KAIST bearing dataset (Jung et al., 2023)	Bearing vibration, motor current and speed reference.	Constant- and variable-speed records; normal, inner-race, outer-race and ball conditions.	Main evaluation: 28 files and 112 tasks in four state groups. Repository Table S1 retains the eight file-index labels. C2 uses SQV.
HUST bearing (Thuan & Hong, 2023)	Vibration; 99 public signals and five bearing types.	Six defect classes and three operating conditions, including compound faults.	Training data and cross-model/compound-fault evaluation.
Paderborn Bearing Data Center (Lessmeier et al., 2016)	Vibration, motor current and rotational speed.	Multiple operating conditions and damage origins.	Training/development and sealed-test data; official identifiers define the partition mapping in Appendix B.8.
SQV (S. Liu et al., 2023)	ch2 vibration (g) and ch3 tachometer pulses (V), 25.6 kHz; 42 of 52 candidate records included.	Seven health states; paired 2 s steady-speed and deceleration windows; 84 tasks in three classes.	Frozen-method evaluation only. Repository Table S15 and Table 8 share full-method runs; seven state groups define the paired resampling.

Huang–Baddour and SQV are reserved for evaluation. Recording membership, window construction and partition checks follow the registered manifests (Appendices B.7–B.8).

Mendeley Data versions: KAIST (Jung et al., 2023), storage Subsets 1–3, version 7 (2023), https://doi.org/10.17632/vxkj334rzv.7, https://doi.org/10.17632/x3vhp8t6hg.7, and https://doi.org/10.17632/j8d8pfkvj2.7, respectively; HUST (Thuan & Hong, 2023), version 3 (2023), https://doi.org/10.17632/cbv7jyx4p9.3.

Online data sources: Paderborn Bearing DataCenter (Lessmeier et al., 2016), https://mb.uni-paderborn.de/en/kat/research/bearing-datacenter; SQV data and speed-extraction code (S. Liu et al., 2023), https://github.com/shenliuuu/SQ-dataset-with-variable-speed-for-fault-diagnosis (both accessed October 15, 2026).

KAIST contributes 28 variable-speed recordings and 112 tasks. Each 300 s vibration recording is paired with its rpm recording and supplies four non-overlapping 60 s windows: [0, 60), [60, 120), [120, 180) and [180, 240) s. Speed is read from the paired rpm file. Repository Table S1 lists K01–K08 file-index labels and public filename aliases. The 240-task manifest records channels, windows, partitions and the source groups used in B.3; storage subsets serve as file organization rather than experimental grouping.

The safety set contains 120 base tasks and 600 paired condition instances. These comprise 120 complete-information controls and 80 instances for each of six interventions. Variants remain grouped with their base task.

B.2 Model and execution settings

Comparable agents share task inputs, the eight registered tools in Repository Table S10, runtime protection, call budgets and timeouts. D-only-full, LLM-process-full and VG-MTP change the offline acceptance rule while matching complete supervision, optimizer updates and packed-sequence capacity. The dynamic rule planner selects paths from intermediate observations using fixed rules. The oracle receives the correct tool family only, without fault labels, geometry, call order or final answers.

Table B.2. Planner training and inference settings.

Item	Frozen value
Backbone and adaptation	Qwen/Qwen3.5-9B; bf16; LoRA rank 16, scaling 32, dropout 0.05.
Training examples	1,024 trajectories per method and seed; matched tasks, physical sources and supervision conditions (A.3 and B.9).
Optimization	AdamW; peak learning rate 1×10^{-4}; 3% linear warmup then cosine decay; 192 updates; effective batch 16.
Sequence and supervision	8,192-token packed sequences; loss on planner calls, visible repairs and reports; tool observations and context masked.
Inference	Context 32,768; output limit 1,024; temperature 0.2; top-p 0.8; repetition penalty 1.05.
Training seeds	11, 23, 37, 53, 71.
Shared execution	Eight tools; canonical v2.0.0; at most 12 dispatched calls; tool/task timeout 30/180 s.
Token accounting	Capacity 25,165,824 token positions per seed; corpus totals and actual consumption are recorded separately (B.9).

Complete software and model revisions, optimizer settings, packing rules and configuration records are provided in docs/configuration.md and training/training_logs/ of the reproducibility repository.

The cost-efficient frontier baseline is gpt-5-mini-2025-08-07. The high-capability baseline and teacher use gpt-5-2025-08-07. Both use the Responses API with medium reasoning effort. The shared registry comprises signal_qc, stft, fft_envelope, order_resample, order_spectrum, bearing_order, claim_checker and expert_classifier, all at v1.0. The auxiliary classifier uses frozen checkpoint exp_resnet1d_trn_v1, fitted only on the Repository Table S6 training partition; its applicability calibration is defined in C.3 and its contract in Repository Table S10.

Each trainable method has $180 \times 3 \times 3 + 60 \times 2 \times 3 = 1{,}980$ primary executions per seed and 9,900 across five seeds. Appendix A's strict-majority rule yields 1,200 task–seed outcomes. Timeouts and executions without a valid terminal state remain in the planned denominator as $\mathrm{EDS} = 0$. Repository Table S14 instead counts base tasks with at least one timeout or budget-exhaustion event. Untrained comparators use $S = 1$.

SQV uses five frozen VG-MTP checkpoints with the settings in Tables B.2–B.3. One prompt and one execution per task, seed and pipeline give 42 records × two phases × five seeds × two pipelines = 840 executions. ALT disables order_resample and order_spectrum, retaining the other six tools without retraining. Both pipelines can produce g_3 localization from mechanistic evidence and apply the same guard to auxiliary expert classification (C.6). Table 8 reuses the 420 full-method executions. Failed outcomes remain in the results without replacement retries.

B.3 Paired statistical analysis

The primary paired analysis joins outcomes on task ID and training seed for the same 240 tasks. Within each of 10,000 bootstrap replicates, it samples four KAIST, six Paderborn and five Huang–Baddour source groups with replacement, retaining all tasks in each sampled group. Within-source task means are combined with fixed weights 112/240, 68/240 and 60/240. Five training seeds are sampled with replacement; untrained methods retain fixed $S = 1$ outcomes. Methods share the same draws. Differences are computed before the 2.5th and 97.5th linear percentiles. The repository records the PCG64 seed 20261013, draw order and implementation in configs/statistics.json and analysis/recompute.py.

The eight-group sensitivity analysis pools KAIST into one group and HB into another, retaining six Paderborn groups. It resamples these groups and five seeds, then computes a pooled task-weighted mean that allows source weights to vary. Table B.5 thus evaluates dependence on both grouping and source composition. The VG-MTP–LLM-process-full interval remains positive; the other VG-MTP contrasts cross zero. Reference-method intervals in Table 4 use their separately specified 14 sealed-test and 20 HB setting groups.

The main paired file contains 1,200 unique task–seed keys, planned repeats, raw success counts and majority outcomes. The reporting ablation uses these same keys and the primary bootstrap. The HB target-prior-withholding comparison uses 20 health-state × speed-setting groups with all three trials retained, sampling

groups and five seeds while keeping Unadapted outcomes fixed. VG-MTP and Unadapted contribute 210/300 and 29/60 successes, respectively.

SQV uses seven state groups and five training seeds. Each of 10,000 replicates samples both dimensions with replacement, retains all six recordings of each sampled group and both phases and pipelines, and averages tasks within seeds and then across seeds. Within-phase pipeline gains and their phase interaction are calculated before percentile intervals. The paired files, source manifests and complete outputs are linked in paper_map.csv.

B.4 Tolerances and perturbation settings

Table B.3. Numerical tolerances and execution limits.

Control	Frozen value	Decision rule
Dispatched-call budget	12	After the twelfth dispatch, only a terminal decision without another tool call is allowed.
Tool / task timeout	30 s / 180 s	Retain timeout as $\mathrm{EDS} = 0$; do not replace a failed execution by a successful retry.
Order matching	0.05 order	Compare theoretical and observed orders before display rounding.
Speed-source conflict	Relative difference > 0.10	An unresolved conflict invalidates dependent evidence.
Tachometer quality	≥ 0.80	Required for speed-synchronous evidence and g_3 localization.
Expert applicability	≥ 0.70	Required for citing expert-model output.
Acceptance score	≥ 0.72	Score only candidates passing the hard evidence gate.
Speed / frequency tolerance	$\max(10\ \mathrm{rpm}, 1\%)$ / $\max(0.5\ \mathrm{Hz}, 1\%)$	Check unit compatibility before numerical agreement.
Score / amplitude tolerance	0.01 absolute / 2% relative	Amplitude comparisons require identical preprocessing and units.

Time is measured in seconds, frequency in hertz and speed in rpm, with explicit unit conversions. Order is dimensionless. Unit compatibility is checked before numerical tolerance. Relative disagreement between speed sources is measured symmetrically.

$$\rho_n = \frac{|n_{\mathrm{meta}} - n_{\mathrm{tach}}|}{\max(n_{\mathrm{meta}}, n_{\mathrm{tach}})}.$$

The acceptance score uses grouped five-fold logistic calibration on the development set with regularization $\lambda = 1$. Its frozen intercept is $b = -2.0$ and weights are (1.0, 0.6, 1.2, 0.8, 0.5, 0.3, −1.5), in the feature order below. The operating threshold is $\theta^* = 0.72$. Score only candidates that pass the evidence gate. REC retains the requested-granularity reference; $\Xi_e(g)$ contains conflicts relevant to the candidate.

$$c_e(g) = \sigma(b + w^{\mathrm{T}} z_e(g)),$$

$$z_e(g) = \left(q_{\mathrm{sig}}, a_{\mathrm{dom}}, \mathrm{REC}, \mathrm{NF}, \mathrm{PC}, p_{\mathrm{plan}}, \mathrm{I}[\Xi_e(g) \neq \emptyset]\right).$$

Band-limited, zero-mean Gaussian noise is added at target SNR values s of 20, 10, 5 and 0 dB. For original in-band signal power P_x, the noise variance and perturbed signal are defined below.

$$\sigma_\eta^2 = P_x \cdot 10^{-s/10}, \qquad x_s(t) = x(t) + \eta_s(t).$$

Tachometer perturbations use r values of 0, 1, 3, 5 and 10%. Each sample is dropped with probability $r/100$. Retained timestamps receive zero-mean jitter as follows.

$$\varepsilon_k \sim \mathcal{N}\left(0, \left[\frac{r}{100} T_{\mathrm{rev}}(t_k)\right]^2\right), \qquad t_k' = t_k + \varepsilon_k,$$

Here $T_{\mathrm{rev}}(t_k) = 60/n(t_k)$ is the local revolution period in seconds, with speed measured in rpm. Perturbation seeds depend on source group, task and training seed. Paired comparisons use identical perturbations. The physical fault label stays fixed, but answerability may change. The task specification determines whether the perturbed input permits diagnosis, a coarser diagnosis or abstention. Perturbation levels are fixed before test outcomes are inspected. For Repository Table S20, the abstention-required reference sets contain 18, 25, 40, 68 and 105 tasks per seed at the original, 20, 10, 5 and 0 dB conditions, respectively. These sets determine the unsafe-acceptance denominators independently of method predictions.

B.5 Training scale and expert audit design

The 1,024-trajectory condition reuses the primary VG-MTP checkpoint for each training seed and the same evaluation results. The 250- and 4,096-trajectory conditions use a separate pool containing only training data; development and test tasks are excluded. This expansion pool is distinct from the 1,280 candidates used in the primary comparison. Table B.4 reports EDS and training cost for all three sizes.

Table B.4. Supplementary training-scale results.

Number of qualified trajectories	Test EDS (%)	Training cost (GPU h)
250	63.3	13.3
1,024	74.50	34.7
4,096	73.1	119.6

Note: The 1,024-trajectory row reuses the primary VG-MTP checkpoints and evaluation results reported in Repository Table S12.

Two mechanical-diagnosis experts independently assess 120 accepted and 120 rejected trajectories while blinded to verifier decisions (Table 7). Report support is assessed separately using 120 anonymized reports per evidence condition, with hidden method labels and randomized order. Expert agreement is calculated before adjudication. Trajectory plausibility is measured per trajectory, while report support is measured per claim; the respective results appear in Table 7 and Repository Table S19.

In the trajectory audit, 113/120 accepted trajectories are expert-plausible (94.2%; two-sided Wilson 95% interval [88.4%, 97.1%]). Among rejected trajectories, 96/120 are implausible (80.0%; [72.0%, 86.2%]) and 24/120 are plausible (20.0%; [13.8%, 28.0%]). Expert–expert Krippendorff's α is 0.82. These estimates describe the two sampled decision strata. Population recall, specificity and false-rejection rates require weighting by the candidate pool's actual accepted and rejected proportions.

For the no-citation, citation-field-only and mandatory-citation-plus-check conditions, same-claim automated UCR is 18.58%, 11.05% and 3.75%, and adjudicated expert UCR is 19.44%, 12.07% and 4.46%. Agreement is 97.05%, 97.62% and 98.57%. Taking expert-unsupported claims as positive, automated recall is 90.18%, 85.92% and 76.00%, and precision is 94.39%, 93.85% and 90.48%. The full-check condition misses 6/25 expert-unsupported claims. Expert–expert Krippendorff's $\alpha = 0.79$ is a separate statistic from automated–adjudicated agreement.

B.6 Results by training seed

Repository Tables S2 and S3 report all five training seeds and link to task-level outcomes and recomputation scripts. Main-evaluation EDS ranges from 72.08% to 76.67% for VG-MTP and from 70.42% to 75.00% for LLM-process-full. The paired VG-MTP gain is positive in every seed, ranging from 1.67 to 2.08 percentage points. Each trained condition contributes 240 outcomes per seed. Unadapted and Dynamic rule are fixed $S = 1$ references with 136/240 and 163/240 successes.

For SQV, each seed contributes 42 binary outcomes per phase and pipeline. The phase × pipeline interaction is positive across all seeds (7.14–11.90 points) and averages 9.52 points. It is computed as (deceleration/full − deceleration/alternative) − (steady/full − steady/alternative) before rounding.

B.7 SQV recording manifest and window protocol

SQV (S. Liu et al., 2023) is used only for frozen-method evaluation. The candidate manifest contains 52 recordings from seven state groups; 42 recordings (six per group) satisfy the same prespecified inclusion rules. Each contributes one 2 s steady-speed task and one 2 s deceleration task, giving 84 tasks and 840 executions across five seeds and two pipelines. Both pipelines share the same recordings, windows, checkpoints and evaluation settings.

Windows use synchronized ch2 vibration (g) and ch3 tachometer pulses (V) at the 25,600 Hz header sampling rate. Pulse edges follow the source convention; speeds are calculated from sample-index differences and smoothed with a centered nine-point median filter. Let p be the maximum filtered speed. The stable segment is the longest continuous interval at or above $0.95p$; its centered 2 s window defines the steady phase. The deceleration window starts at the segment's final speed timestamp, must decrease by at least $0.10p$ over 2 s, and permits no successive filtered-speed increase above 5 rpm.

Both complete windows must fit the recording and contain 51,200 samples per channel. Nine candidate recordings lack a qualifying stable segment; one lasts 1.48 s and has no valid tachometer pulses. No padding, concatenation or synthetic extension is used. Inclusion and window selection were fixed before diagnostic evaluation and applied uniformly; SQV was not used for method or threshold selection.

Repository Tables S4 and S5 map local record IDs to public files and give the actual start times, rounded to 0.01 s, for the half-open windows [start, start + 2 s). The machine-readable manifest preserves all candidate records, inclusion decisions and window boundaries. docs/protocols/sqv_windows.md gives the pulse-edge convention and complete selection protocol. NSK (n.d.) provides nominal specifications for the 6203 bearing. The repository training configurations record the fixed geometry used to calculate theoretical orders: eight rolling elements, rolling-element diameter 6.75 mm, pitch diameter 29.0 mm and contact angle 0°. These parameters give $\mathrm{BPFI} = 4.93103$ and $\mathrm{BPFO} = 3.06897$ orders.

B.8 Main evaluation mapping and partition audit

Repository Table S6 registers partition selections after Paderborn alias normalization; Repository Table S7 maps the main-evaluation groups to tasks. Training uses 168 recording keys (99 HUST and 69 Paderborn), and development uses 56 Paderborn records. The main evaluation uses 28 KAIST, 68 Paderborn and 60 Huang–Baddour recordings, yielding 112, 68 and 60 tasks, respectively. KAIST vibration and speed files are modalities of the same recording.

The nine training, 14 development and six sealed-test Paderborn identifiers are disjoint. Canonicalized recording-key sets also have zero pairwise intersections across partitions. Each key combines the dataset name and recording ID. Registered source groups preserve within-source dependence; physical-specimen independence is not assumed for every group.

The 240-task manifest supplies file, channel, window, group and partition for each task. KAIST uses four 60 s windows per recording; Paderborn and HB use complete 4 s and 10 s records, respectively. There are no duplicate or missing task–seed keys in the paired main results. Each trained method has 9,900 planned executions and 1,200 majority outcomes. VG-MTP, LLM-process-full, D-only-full and Final-label-only have 894, 873, 828 and 785 successes; fixed references have 136/240 and 163/240 successes.

Table B.5. Sensitivity of paired EDS gains to source grouping.

Paired comparison	Gain (pp)	15-group interval (pp)	8-group sensitivity (pp)
VG-MTP − Unadapted	+17.83	[15.26, 20.06]	[−14.26, 27.56]
VG-MTP − Final-label-only	+9.08	[6.72, 11.43]	[−23.72, 20.92]
VG-MTP − D-only-full	+5.50	[3.78, 6.92]	[−28.22, 18.33]
VG-MTP − LLM-process-full	+1.75	[1.00, 2.50]	[0.63, 3.19]
VG-MTP − Dynamic rule	+6.58	[4.50, 8.45]	[−28.44, 18.83]
LLM-process-full − D-only-full	+3.75	[1.75, 5.40]	[−30.00, 17.11]

Note: All rows use the same task keys and 10,000 paired resamples. The 15-group analysis fixes source weights; the eight-group analysis allows them to vary (B.3). C1a and C1b meet their criteria only in the 15-group analysis. VG-MTP versus LLM-process-full remains positive under both schemes.

B.9 Training sources and process filtering control

The matched control applies three offline acceptance rules to the same 1,280-candidate pool, backbone, tool environment and complete-trajectory supervision. The source register links candidates to 252 training tasks, 504 input entries and 168 recording keys (99 HUST and 69 Paderborn), all in the training partition. Each final corpus contains 1,024 parent trajectories matched on task, physical source, fault class, speed regime, path length and teacher source.

The repository provides candidate-to-task, task-to-input and corpus-to-candidate mappings, including windows, channels and registered physical-file hashes. Paderborn vibration uses Y(Name = vibration_1).Data and its X time axis. Nominal-speed inputs are identified separately from the available Y(Name = speed).Data channel; the field mapping preserves the original corpus content.

After the shared terminal check D, Qwen3.5-32B-Instruct, DeepSeek-V3 and GPT-4o-mini review each complete visible trajectory. The fixed prompt supplies the same tool contracts, physical conditions, units, tolerances, dependency rules and claim–evidence requirements available to the deterministic verifier, together with permitted task metadata, actions, actual observations, visible purpose codes, repairs and the terminal report. Offline verifier labels, selection flags, test answers and private teacher reasoning are excluded. Each reviewer uses temperature 0 and a maximum output of 1,024 tokens.

Reviewers assess each dispatched call's physical prerequisites, the validity and order of upstream states, and the evidence supporting the diagnosis and granularity. Each returns pass/fail with reasons. Eligibility requires D and unanimous approval from all three reviewers. A timeout, truncated response or invalid format permits one retry; a second technical failure excludes the candidate and is counted separately from rejection. The control extends TimeART's sampled-chain review (Wu et al., 2026, Section 2.2) to the complete visible trajectory while retaining unanimous approval.

Table B.6. Candidate screening and matched training counts.

Condition	Common candidates	D pass	Eligible before matching	Final training
D-only-full	1,280	1,096	1,096	1,024
LLM-process-full	1,280	1,096	1,063	1,024
VG-MTP	1,280	1,096	1,030	1,024

Repository Table S8 gives the complete matching strata. Stratum A uses 84 records and selects 478 candidates; B uses 33 HUST records and selects 256; C uses 51 Paderborn records and selects 290. The recording sets are disjoint. Candidate-pool labels such as HB and KAIST are quota templates; dataset and record_key identify the physical training inputs. Evaluation HB denotes Huang–Baddour.

Of the 1,096 D-passing candidates, the LLM panel accepts 1,063, rejects 30 and leaves three technical failures unresolved. Each of 20 method–seed corpus files contains 1,024 distinct parent candidates and 3,072 instances over three nominal epochs, indexed by corpus_file, epoch and sample_id. All methods use 192 optimizer updates. Nominal corpus-token totals differ by at most 0.181% among matched full-trajectory conditions. Actual consumption is recorded separately: for seed 37, LLM/VG totals are 23,282,400/23,246,800, a 0.153% difference, both below the shared capacity of 25,165,824. Per-seed consumption, mappings and screening records are provided in the training logs and accompanying files.

B.10 Runtime and reporting ablations

Runtime and reporting ablations change one component at a time under the shared execution settings (Table 9 and Repository Table S18). The full/no-binding comparison contains 1,200 task–seed pairs: 826 succeed under both, 68 under full only, none under no-binding only and 306 under neither. Full EDS is 74.50%, compared with 68.83% without mandatory binding; the full-minus-no-binding gain is 5.67 percentage points (95% interval [4.55, 6.61]) under the primary paired bootstrap.

The repository supplies paired outcomes and their configuration correspondence. The full condition reuses the primary evaluation; other module differences use the displayed point estimates. Each ablation estimates a component's removal effect with the remaining components retained. Effects are evaluated separately, and every reporting condition retains G's factual-source requirement.

Appendix C Execution Rules and Call Audit

This appendix specifies evidence predicates, state and conflict handling, acceptance scoring, planning procedures and tool parameters. Runtime checks and offline physical review use full-precision features and complete dependency states at the time of each call; offline terminal scoring additionally uses reference labels. Table C.3 provides the complete speed-conflict trajectory, and Table C.4 summarizes call counts and agreement between checks.

C.1 Required evidence by diagnostic granularity

Fault diagnosis comprises anomaly detection (g_1), bearing-fault diagnosis (g_2) and component localization (g_3); normal conditions use a separate health label. g_2 requires E1 and E2, while g_3 additionally requires E3. Required predicates are evaluated from actual tool outputs, and key claims undergo provenance, numerical, semantic and dependency checks. The requested granularity g_{req} is fixed, g_{try} is the current candidate granularity, and $g_{\min}$ is the minimum granularity permitted by the task. Downgrading updates g_{try} and reevaluates evidence at the new candidate granularity.

Table C.1. Evidence predicates used by the frozen evaluator.

ID	Applicable conclusion	Criterion	Tool and tolerance
E1.1	g_1–g_3	Sampling rate matches the declared value; no clipping; all required channels are present.	signal_qc; normalized-score tolerance 0.01.
E1.2	g_1–g_3	At least one holds: kurtosis $>$ 3.5, crest factor $>$ 4.5 or broadband-energy change relative to the reference $\geq$ 3 dB.	signal_qc or spectral recomputation.
E2.1	g_2–g_3	A significant envelope-spectrum peak occurs in a bearing-characteristic band; prominence $\geq$ median + 6σ.	fft_envelope; frequency tolerance max(0.5 Hz, 1%).
E2.2	g_2–g_3	Impact periodicity, envelope-peak spacing or sideband spacing agrees with a bearing-characteristic frequency; the associated speed source is reliable.	fft_envelope; frequency tolerance max(0.5 Hz, 1%).
E3.1	g_3	Rolling-element count, rolling-element diameter, pitch diameter and contact angle are complete and pass geometry-validity checks.	bearing_order prerequisite checks.
E3.2	g_3	Theoretical BPFO, BPFI and BSF orders have been calculated.	bearing_order.
E3.3	g_3	At least one match is established: order-spectrum, constant-speed frequency or variable-speed ridge-based instantaneous order.	Order deviation $\leq$ 0.05 order; frequency tolerance max(0.5 Hz, 1%).
E3.4	All g_3 paths	$\text{tach}_q \geq 0.80$, with no unresolved conflict affecting the speed source used.	signal_qc and verifier.
N1	health	Adequate signal quality, as in E1.1.	signal_qc.
N2	health	All three hold: kurtosis $\leq$ 3.5, crest factor $\leq$ 4.5 and broadband-energy change $<$3 dB.	Same statistics and reference as E1.2.
N3	health	No significant envelope-spectrum peak under the E2.1 criterion.	fft_envelope.
N4	health	No significant observed peak near a theoretical fault feature.	bearing_order and the relevant spectral tool; E3.3 tolerances apply.

E3.3 accepts three types of matching evidence. The full pipeline matches orders observed by order_spectrum against theoretical orders from bearing_order. The constant-speed path matches envelope-spectrum peaks against frequencies obtained by multiplying theoretical orders by shaft frequency. The variable-speed alternative converts stft ridges to orders using $o(t) = 60f(t)/n(t)$ and compares the within-window median order with the theoretical value. Constant speed requires the range of smoothed speed within a window to remain within $\max(10\,\text{rpm}, 1\%)$, with the relative term referenced to that window's speed estimate. A normal diagnosis requires N1–N4 jointly.

An unresolved SPEED_SOURCE_CONFLICT invalidates E2.2, E3.3 and E3.4 when they depend on the disputed source. After downgrading, required evidence and dependencies are recalculated for the new candidate; all predicates remain grounded in actual observations.

C.2 Evidence state and conflict handling

The evidence state is generated deterministically from the append-only ledger L, task specification and current candidate granularity. Verified metadata M come from valid observations, candidate hypotheses H correspond to the current granularity, and supporting and contradictory evidence E^+/E^- retain the mapping claim_id $\rightarrow$ (call_id, field, value) between claims and call fields. Gaps N list unmet required predicates, while Ξ stores unresolved conflicts. The executor updates remaining calls and wall-clock time, supplying the budget input b to Eq. (8). Identical ledger, specification, candidate-granularity and budget inputs yield identical states.

The ledger retains execution states, raw outputs, versions and provenance for actual calls. Error records support auditing, while only valid observations can support claims. Blocked proposals enter the execution trace, and structured repair feedback enters the planner context ctx; neither creates a tool observation nor consumes a dispatch slot. Corrections are appended as new versions, preserving the original records.

Table C.2. Claim semantics and allowed evidence fields.

Claim type	Permitted tools and fields	Verification rule
peak_observation	Peak frequency or order, amplitude and prominence from stft, fft_envelope or order_spectrum	Recompute from the corresponding raw output; check units before applying tolerances.
theoretical_order	BPFO, BPFI, BSF and FTF from bearing_order	Recompute from verified geometry inputs.
order_match	match_deviation from bearing_order	Verify observed and theoretical sources; order deviation ≤ 0.05.
speed_estimate	Speed series or estimate from signal_qc	Recompute and check the $\max(10\,\text{rpm}, 1\%)$ tolerance and source conflicts.
diagnosis(g)	Combined claims for the corresponding granularity	All $E(g)$ predicates hold, and key claims pass all four checks.
health	Combined claims for a normal condition	N1–N4 all hold, and key claims pass all four checks.

P1 denotes high-priority conflicts, including inconsistent speed sources and missing or invalid required geometry. Speed conflicts use $\rho = \frac{|n_{\text{meta}} - n_{\text{tach}}|}{\max(n_{\text{meta}}, n_{\text{tach}})} > 0.10$. A P1 conflict blocks calls and claims that depend on the disputed quantity. The gate checks the current candidate's dependency chain; downgrading does not delete conflict records from the ledger.

P2 denotes disagreement among redundant indicators, such as abnormal statistics without a significant envelope peak. It contradicts directly related claims, while candidates independent of those claims can still undergo evidence gating. The set $\Xi(g)$ retains conflicts relevant to the candidate, and the scoring term reflects relevant P2 conflicts. Under the fixed coefficients in C.3, an unresolved relevant P2 limits the acceptance score to 0.711, below 0.72. The system must resolve the conflict or consider a valid candidate independent of that evidence before rescoring.

The evidence gate $G(S, g)$ requires complete essential evidence, four-check validity of key claims, compliance with requested granularity and no unresolved P1 conflict relevant to the current candidate. A candidate that passes this gate must also meet the acceptance-score threshold.

C.3 Acceptance scoring from quality and applicability

Tachometer quality is $\mathrm{tach}_q = \frac{q_{\mathrm{edge}}+q_{\mathrm{regularity}}+q_{\mathrm{agreement}}}{3}$, with each component clipped to [0,1]. Edge completeness is $q_{\mathrm{edge}} = \min\left(1, \frac{N_{\mathrm{valid\ edges}}}{N_{\mathrm{expected}}}\right)$, where $N_{\mathrm{expected}} = T \cdot n_{\mathrm{meta}}/60$. If no speed is declared, the median speed derived from valid pulse-edge intervals determines the expected pulse count. Regularity is $q_{\mathrm{regularity}} = 1 - \mathrm{CV}(\Delta T_{\mathrm{edge}})$, where ΔT_{edge} denotes successive valid edge intervals. When two speed sources are available, agreement is $q_{\mathrm{agreement}} = 1 - \min(1, \rho)$. Quality and source consistency are checked separately: $\mathrm{tach}_q \geq 0.80$ and $\rho > 0.10$ can hold simultaneously, and the latter still triggers conflict handling.

Signal quality is $q_{\mathrm{sig}} = \frac{1-\mathrm{clip}_{\mathrm{frac}}+\mathrm{clip}(\mathrm{SNR}_{\mathrm{dB}}/40,0,1)+f_{s,\mathrm{ok}}}{3}$, where $\mathrm{clip}_{\mathrm{frac}}$ is the clipped-sample fraction and $f_{s,\mathrm{ok}}$ indicates agreement between the measured and declared sampling rates. Negative SNR values contribute zero after normalization. Repository Table S9 specifies the power estimates used for SNR.

Applicability features comprise normalized energies in 12 one-third-octave bands, spectral kurtosis, RMS magnitude, one-hot sensor modality, sampling-rate bins and mounting-orientation categories. Mahalanobis distance d is computed using the training-set mean and covariance, both fixed before evaluation. The score is $a_{\mathrm{dom}} = 1 - \min(1, d/d_{\mathrm{ref}})$, where d_{ref} is the 95th percentile of training-set distances. Expert-model outputs can be cited in a terminal conclusion only when $a_{\mathrm{dom}} \geq 0.70$. If no expert-model output is cited, the applicability feature is set to 1.

The planning-confidence feature p_{plan} is the planner action head's softmax probability, at the fixed temperature, for the selected terminal type ACCEPT(g), DOWNGRADE or ABSTAIN. The candidate and its probability are stored before evidence gating and acceptance scoring. Offline labels and final success outcomes do not enter this calculation. Model and decoding settings follow Table B.2.

REC, NF and PC are calculated within each execution and used as scoring features. REC uses the required evidence set for the fixed requested granularity g_{req}, whereas the candidate gate uses $E(g_{\mathrm{try}})$. A coarser candidate can therefore pass its own gate while retaining REC below 1. NF is the proportion of numerical claims that pass recomputation, and PC is the proportion of citation-requiring claims with traceable sources. Global reporting pools applicable numerators and denominators across runs and seeds.

Acceptance scoring uses grouped five-fold logistic calibration on the development set with regularization $\lambda = 1$. The fixed intercept is $b = -2.0$, with coefficients $w = (1.0,0.6,1.2,0.8,0.5,0.3,-1.5)$ corresponding to q_{sig}, a_{dom}, REC, NF, PC, p_{plan} and $\mathrm{I}[\Xi(g) \neq \emptyset]$, respectively. The score is $c = \sigma(b + w^{\mathrm{T}}z)$, with $\sigma(a) = \frac{1}{1+\exp(-a)}$. The main threshold $\theta^* = 0.72$ is fixed at the development-set selective-F1 operating point. Only candidates passing the evidence gate are scored. A gate failure leads to evaluation of a permitted coarser candidate or evidential abstention.

For example, a candidate that passes the evidence gate with $z = (0.95,1,1,1,1,0.85,0)$ receives $c = \sigma(2.305) \approx 0.909$ and can be accepted. With an unresolved relevant P2 conflict, even unit values for the other six features yield at most $\sigma(-2 + 4.4 - 1.5) = \sigma(0.9) \approx 0.711$, below the acceptance threshold, causing selective abstention. Test-time risk–coverage analysis sweeps thresholds over the fixed score while retaining the evidence gate.

C.4 Main loop and terminal decisions

The main loop follows the sequence below. Complete pseudocode is provided in docs/method/planner_pseudocode.md in the accompanying repository. The requested granularity g_{req} remains fixed; g_{try} denotes the current candidate granularity.

1. Initialize the append-only ledger L, execution trace, repair context, disabled-call set and budget. Generate the evidence state from valid observations, the task specification, candidate granularity and remaining budget.

2. The planner proposes a call, downgrade, stop or abstention. Check the 180 s total limit at each iteration and after the planner returns. Before a call, check its schema, units, dependencies and applicability. Blocked proposals record feedback only, producing no observation and consuming no dispatch slot. A previously blocked call cannot be dispatched again without a substantive repair.

3. Execute valid calls within the 12-dispatch budget, with a 30 s limit per call, and append outputs and provenance to the ledger. An unrecoverable error returns execution failure. A timeout records one structured feedback item and returns budget failure. Both remain in the EDS denominator. After the twelfth call, a terminal decision requiring no additional call is still permitted.

4. Downgrading is permitted only when $g_{\min} \leq g < g_{\text{try}}$, with evidence and dependencies recalculated accordingly. Normal diagnoses require N1–N4. Other candidates are evaluated from the current granularity toward permitted coarser levels.

5. Compute the fixed acceptance score only after the candidate passes checks of required evidence, key claims, requested granularity and relevant P1 conflicts. Accept when $c \geq 0.72$; return selective abstention when the evidence gate passes but the score is insufficient.

6. If no permitted candidate passes the evidence gate, return evidential abstention with confirmed facts and reasons. Total timeout and unrecoverable errors take precedence over acceptance.

C.5 Dynamic rule-based planner

The dynamic rule-based planner has no learned parameters, and its branches are fixed before evaluation. It shares inputs, tools, runtime verification, dispatch budgets and timeouts with VG-MTP. Repair feedback takes precedence over routine path selection. Disabled tools and their dependencies remain disabled until the execution ends, and every call-dispatch route applies the same check.

A speed-source conflict triggers one recheck. If unresolved, calls dependent on the disputed source are skipped. Tachometer-quality or applicability failures likewise disable the affected tools and dependencies. Missing geometry prompts evaluation of coarser conclusions within the permitted range. After the initial signal_qc call, the planner uses a spectral path if no reliable speed is available. With reliable speed, it uses an order-domain path when within-window variation exceeds $\max(10\ \text{rpm}, 1\%)$ and a spectral path otherwise.

The order-domain path obtains missing envelope evidence, revalidates speed, performs angular resampling, computes the order spectrum, matches theoretical features and checks claims. The spectral path adds STFT or theoretical matching according to evidence gaps. Normal candidates also require N4. Planning stops when evidence is sufficient; otherwise, it attempts a valid downgrade or abstains. It also stops when no calls remain or remaining time is below 30 s.

Mean call counts were 6.0 for the dynamic rule-based planner and 6.49 for VG-MTP, with EDS values of 67.92% and 74.50%, respectively (Table 5). Sensitivity to source grouping follows the analysis in Table B.5.

C.6 Fixed parameters for the full and alternative pipelines

COMPLETE uses eight tools: signal_qc, stft, fft_envelope, order_resample, order_spectrum, bearing_order, claim_checker and expert_classifier. ALT disables order_resample and order_spectrum while retaining the other six tools and all applicability safeguards. Both pipelines can perform g_3 localization. They may use frequency matching under steady speed. Under variable speed, COMPLETE uses angular resampling and order spectra, whereas ALT converts STFT ridges to instantaneous orders using reliable speed information. SQV inputs from a different test rig undergo identical applicability checks; mechanism-based evidence supports g_3, with expert probabilities serving as auxiliary information. The comparison tests access to the two order-analysis tools, with speed information available in both conditions.

Both pipelines use a 2,560-sample Hann window with 75% overlap for STFT and a squared Hilbert-envelope spectrum with FFT length 16,384. The full pipeline resamples at 128 points per revolution, while the

alternative converts instantaneous orders using $o(t) = 60f(t)/n(t)$. Theoretical matching retains the 0.05-order or $\max(0.5\,\text{Hz}, 1\%)$ tolerance.

Both pipelines share the tolerances and execution limits in Table B.3, as well as tool versions, prompts, paired seeds and terminal rules. Parameters and scoring rules were fixed during training and development; SQV did not contribute to their selection.

C.7 Tool contracts and call examples

All eight tools use v1.0 and expose identical interfaces to comparable agents. Each dispatch records inputs, actual outputs, units, prerequisites, error states, tolerances, versions, provenance hashes and costs. Repository Table S10 specifies individual contracts; docs/method/tool_contracts.md provides geometry inputs, theoretical fault-order equations and error-handling examples.

Tool dependencies are checked before dispatch. order_spectrum requires successful angular resampling; bearing_order requires valid geometry and conflict-free speed; and expert_classifier requires successful signal_qc checks and $a_{\text{dom}} \geq 0.70$. The applicability score is independent of classification confidence, and g_3 localization still requires mechanism-based evidence. Key claims can support a conclusion only after passing provenance, numerical, semantic and dependency checks.

Blocked proposals enter the execution trace but produce no citable observation. The log name expert_cls is an alias for expert_classifier; each call_id counts as one dispatch. The speed-conflict example in Table C.3 links prerequisite checking, call blocking and terminal abstention.

C.8 Complete trajectory under conflicting speed information

The safety-evaluation task case_conflict_042 contains conflicting speed information. Inputs include a vibration window and tachometer channel at $f_s = 25{,}600\,\text{Hz}$, file metadata specifying 1,480 rpm and complete geometry fields. The request seeks BPFI localization at g_3, with minimum permitted granularity g_2. The specification permits abstention if the conflict cannot be resolved.

Table C.3. Executed trajectory and blocked proposal for case_conflict_042.

Step	Action and execution status	Observation or verification result
1	Initial signal_qc check; proposed and dispatched	$q_{\text{sig}} = 0.98$; $\text{tach}_q = 0.92$; tachometer estimate 1,812 rpm. $\rho = 332/1812 \approx 0.183 > 0.10$; SPEED_SOURCE_CONFLICT recorded.
2	bearing_order using file speed; proposed and blocked	Speed-source conflict remains unresolved; VERIFY_PRE returns SPEED_SOURCE_CONFLICT. No observation is produced and no dispatch slot is consumed. Feedback requests source revalidation or reduced claim granularity.
3	signal_qc speed recheck; proposed and dispatched	Estimate remains 1,812 rpm, with $\text{tach}_q = 0.93$; the source conflict persists.
4	Check task payload for a third speed source; internal action	No additional source resolves the conflict. This is not a tool call, and the call from step 2 is not reproposed unchanged.
5	fft_envelope; proposed and dispatched	Impact evidence is confirmed, but the reliable shaft frequency required by E2.2 remains unavailable; a g_2 bearing-fault diagnosis is unsupported.
6	Check downgrade boundary; internal decision	Anomaly evidence at g_1 exists, but g_1 is below the required $g_{\min} = g_2$ and cannot be issued as an admissible diagnosis.

Step	Action and execution status	Observation or verification result
7	Evidential abstention $\perp_{\text{evid}}$	Report retains confirmed impacts, the unresolved speed conflict and a recommendation to repeat measurement, without a scorable fault label.

In step 1, $q_{\text{edge}} = 1$, $q_{\text{regularity}} = 0.95$ and $q_{\text{agreement}} \approx 0.82$; in step 3, $q_{\text{regularity}} = 0.96$, yielding quality scores of approximately 0.92 and 0.93. Both quality checks pass, while the P1 source conflict persists and restricts dependent calls.

The trajectory contains four call proposals, one block and three dispatches, consuming 3 of 12 slots. Cited observations come from steps 1, 3 and 5. None of these dispatched calls depends on the disputed speed, yielding a TCPR count of 3/3. The task permits abstention, cited observations originate from actual execution, prerequisites and budget and action limits are satisfied, and the report preserves confirmed facts and evidence gaps. The run therefore passes $D/X/P/T/G$ and receives $\text{EDS} = 1$.

C.9 Call accounting and consistency checks

A structured CALL generated by the planner is first recorded as a proposal. Proposals that fail pre-dispatch checks enter the trace; dispatched calls produce execution records and contribute to call counts. TCPR uses dispatched calls with applicable prerequisites, with compliance determined by offline review. Report acceptance additionally requires the candidate evidence gate, acceptance score and claim checks; final diagnostic outcomes are scored separately by the offline evaluator.

Table C.4. Call counts and validator consistency over 9,900 primary VG-MTP executions.

Statistic	Exact count	Counting rule
Raw runs	9,900	Total across five seeds; failures remain in the planned denominator.
Proposed CALLs	72,418	64,210 dispatched + 8,198 blocked + 10 incomplete decisions.
Blocked CALLs	8,198	Not dispatched; excluded from the TCPR denominator.
Incomplete dispatch decisions	10	Counted neither as dispatches nor as agreed passes.
Dispatched calls / applicable denominator	64,210	Includes applicable calls dispatched during failed runs.
Calls satisfying shared prerequisites	64,210	$\text{TCPR} = 64{,}210/64{,}210 = 100\%$.
Comparable proposals	72,408	Both runtime and offline decisions are available for dispatched and blocked proposals.
Runtime/offline disagreements	0	Counted across all comparable proposals.
Comparison coverage	99.986%	72,408/72,418; incomplete decisions are retained separately.

The gate P checks each dispatched call. Any prerequisite violation gives the corresponding raw run $P = 0$ and $\text{EDS} = 0$, regardless of whether the report cites that call. Blocked proposals are treated as undispatched. Terminal correctness, actual execution, trajectory validity and evidence sufficiency must independently pass their respective gates.

Semantic support is assessed by claim–evidence gating and UCR; NF measures numerical faithfulness, and TCPR measures call-prerequisite compliance. In case_ood_accept_006, calls satisfied their prerequisites, but authentic outputs did not adequately support the diagnosis, constituting a semantic-support failure. Safety evaluation scores actual acceptance and abstention outcomes, including erroneous acceptances missed by runtime checks.

Physical features are generated and evaluated at the same call-time state. $q_{\text{regularity}}$ uses unsmoothed pulse intervals, Δn uses the specified nine-point median processing, spectral kurtosis retains its raw value, numerical precision is float64, and dependency validity propagates through all upstream relationships. Training-trajectory filtering, runtime checks and offline physical review use the same specification. State summaries are for presentation; decisions use the original features.

C.10 Paired outcomes for VG-MTP and LLM-process-full

Among 1,200 paired task–seed outcomes, both methods succeed in 869 cases, only VG-MTP succeeds in 25, only LLM-process-full succeeds in four, and both fail in 302. Their strict-majority success counts are 894 and 873, respectively, yielding a net gain of 1.75 percentage points. Repository Table S11 provides per-seed contingency tables, and results/main/paired_task_seed.csv contains all paired outcomes.

The paired gain is positive for every training seed, with 25 VG-MTP-only and four LLM-only successes overall. The records associate these differences with rechecking budgets, acceptance boundaries, claim binding and path switching after applicability failures, providing context for the observed outcomes. The causal contributions of individual checks have not been isolated.

C.11 Equivalent evidence paths and threshold checks

E3.3 accepts matching paths according to their evidence and error. Each pair of rows in Table C.5 shares the input window, g_3 request and geometry checks; alternative paths use frequency or STFT evidence. Steady-speed matching uses the $\max(0.5\,\text{Hz}, 1\%)$ tolerance, while variable-speed ridges use the 0.05-order tolerance after conversion to instantaneous orders. Every path must also satisfy E3.1, E3.2 and E3.4.

Table C.5. Matched-input checks of equivalent evidence paths under the same G rule.

Task	Evidence path	Theoretical	Observed	Deviation / limit	tach_q	G/D
SQV010_S	Order spectrum	4.931 order	4.921 order	$0.010 \le 0.05$	0.96	1 / 1
SQV010_S	Steady frequency	241.37 Hz	240.85 Hz	$0.52 \le 2.414$ Hz	0.96	1 / 1
SQV035_V	Order spectrum	3.069 order	3.048 order	$0.021 \le 0.05$	0.93	1 / 1
SQV035_V	STFT ridge	3.069 order	3.106 order	$0.037 \le 0.05$	0.93	1 / 1
KAIST ball_3	Order spectrum	1.984 order	2.048 order	$0.064 > 0.05$	0.91	0 / 0
KAIST ball_3	STFT ridge	1.984 order	2.112 order	$0.128 > 0.05$	0.91	0 / 0

Note: The SQV windows are IF_1/REC3597 [7.57, 9.57) s and OF_1/REC3513 [11.93, 13.93) s; KAIST ball_3 uses [120, 180) s. The SQV reference fault orders are $\text{BPFI} = 4.93103$ and $\text{BPFO} = 3.06897$. SQV010_S has nine-point median-filtered speed 2,936.90 rpm (48.948 Hz), giving theoretical BPFI 241.366 Hz and a 1% tolerance of 2.41366 Hz. Threshold checks precede display rounding. Paired rows check evidence paths and are not additional independent tasks.

In Table C.5, both steady-speed frequency evidence and variable-speed STFT evidence pass G, whereas both negative-control paths exceed the 0.05-order tolerance. Segmented FFT follows the same E3.3 evidence and tolerance rules. The common rules thus admit different computational paths, while diagnostic accuracy, Macro-F1, coverage and unsafe acceptance assess their diagnostic quality.

C.12 Signal-analysis tools and physical relationships

The evaluated task combines condition-dependent analyses with explicit physical relationships. Statistical tests of hypothesized orders (G. Zhang et al., 2025) and tacholess order tracking (C.

Wang et al., 2026) provide established ways to extract diagnostic evidence. Rotational-frequency estimation (B. Zhang et al., 2025), adaptive ridge extraction (X. Chen et al., 2026) and on-rotor sensing with order tracking (Li et al., 2026) offer additional analysis paths. Their prerequisites make the application suitable for testing whether a planner selects a valid path under the available information.

Open-set and uncertainty-aware models address complementary decisions through evidential reasoning (Fu et al., 2026), frequency awareness (Yu Gao et al., 2025), counterfactual domain learning (S. Liu et al., 2025) and adversarial adaptation (Lei et al., 2025). VG-MTP uses auxiliary predictions within the same contract and evidence requirements as other observations. The experiments therefore retain diagnostic labels and metrics while evaluating action selection, evidence use and conclusion acceptance.

Speed variations shift rotation-synchronous components along the frequency axis and can broaden their energy in a spectrum computed over an entire record. Once the angular-speed trajectory $\omega(t)$ has passed continuity and reliability checks, integration yields angular position and instantaneous rotational frequency:

$$\theta(t) = \int_0^t \omega\,(\xi)\,\mathrm{d}\xi, \qquad f_r(t) = \frac{\omega(t)}{2\pi} \tag{C.1}$$

Resampling at equal angular increments aligns speed-dependent, rotation-synchronous components in the order domain. The instantaneous order corresponding to local frequency $f(t)$ is

$$O(t) = \frac{f(t)}{f_r(t)} \tag{C.2}$$

Theoretical fault orders can be calculated once bearing geometry passes completeness and validity checks. Given valid values for the number of rolling elements n, rolling-element diameter d, pitch diameter D and contact angle φ, the theoretical outer- and inner-race fault orders are

$$O_{\mathrm{BPFO}} = \frac{n}{2}\left(1 - \frac{d}{D}\cos\varphi\right) \tag{C.3}$$

$$O_{\mathrm{BPFI}} = \frac{n}{2}\left(1 + \frac{d}{D}\cos\varphi\right) \tag{C.4}$$

Theoretical and observed orders are matched using fixed tolerances that accommodate deviations caused by slip, load variation and measurement error. Angular-resampling and order-matching tools return prerequisite failures when speed trajectories contain prolonged gaps, nonphysical jumps or severe synchronization errors. Under constant speed, speed-synchronous analysis can be omitted if ordinary spectral or envelope analysis already provides sufficient evidence.

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